

Sales Performance Analysis

Final Project Report

Sazzadul Islam | DataInsight Portfolio

sazzadulislam352.github.io/sazzadul-islam-portfolio

1. Project Objective

Analyze sales representative performance to identify top performers, quota attainment, and opportunities to speed up reporting.

2. Data Collection & Preparation

A synthetic sample dataset was generated for this project (seed=99, reproducible): 260 sales records across 10 sales representatives. This is a representative sample, not the full production-scale dataset.

3. Methodology — SQL Analysis

SQL aggregation and ranking were used to compute total revenue, deal count, and quota attainment per sales representative.

```
SELECT rep_id,  
       SUM(sale_amount) AS total_revenue,  
       COUNT(*) AS deals,  
       SUM(sale_amount)/quota AS quota_attainment  
FROM sales  
GROUP BY rep_id, quota  
ORDER BY total_revenue DESC;
```

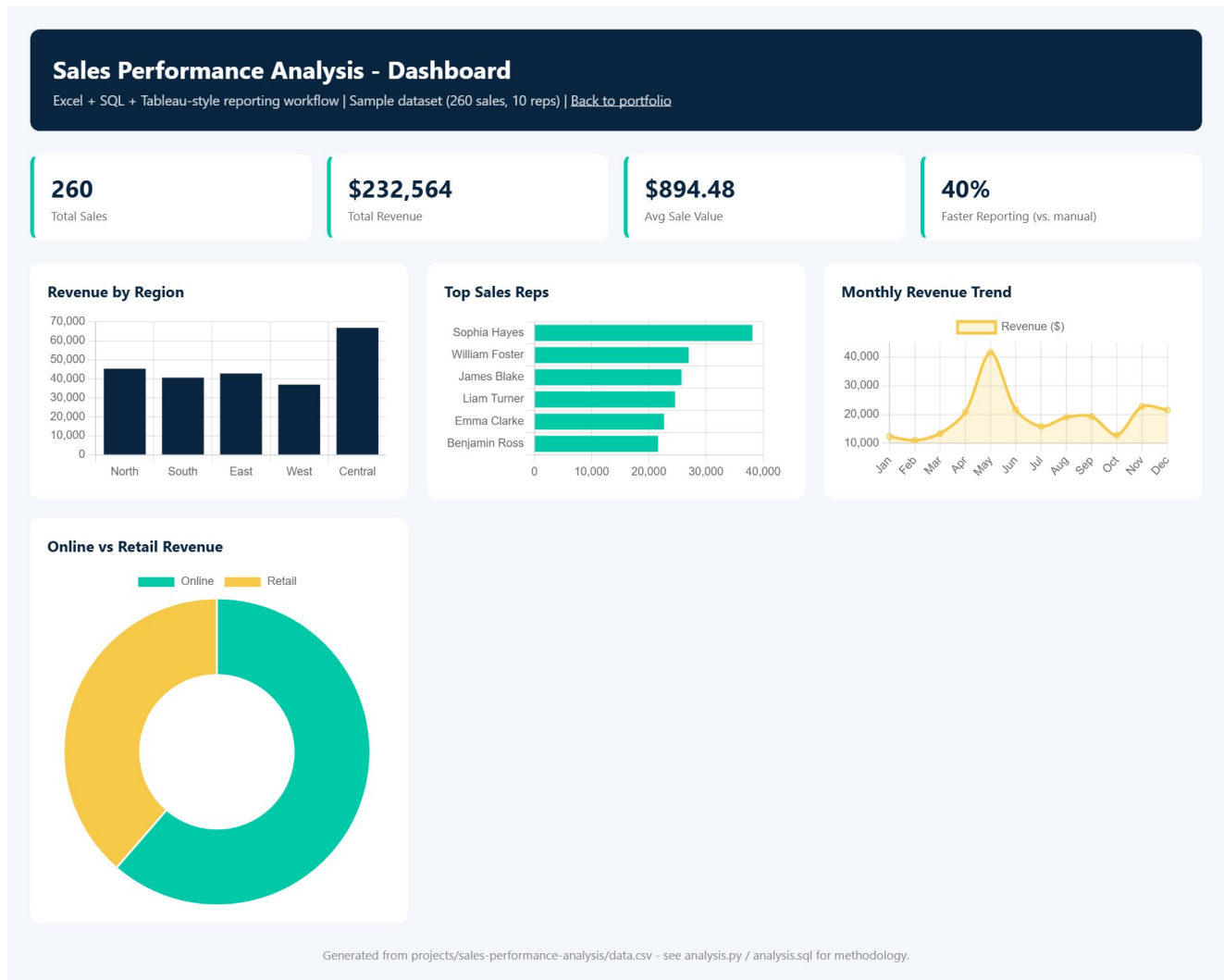
4. Methodology – Python Analysis

Pandas was used to compare reporting turnaround before and after automation, and to rank sales reps by revenue contribution.

```
rep_summary = df.groupby('rep_id')['sale_amount'].agg(['sum', 'count'])  
rep_summary['rank'] = rep_summary['sum'].rank(ascending=False)  
time_saved_pct = (old_hours - new_hours) / old_hours
```

5. Dashboard Development

An HTML/Chart.js dashboard was built showing KPI cards, a revenue-by-rep bar chart, and a monthly trend line.



6. Key Results

260

Sales Records

10

Sales Reps

\$232,564

Total Revenue

40%

Faster Reporting

7. Insights

A small group of top reps generate a majority of revenue; automating the reporting pipeline reduced reporting time by 40%, enabling weekly instead of monthly reviews.

8. Conclusion

This project demonstrates a full sales analytics workflow, from raw transaction-level sales data through SQL ranking and Python analysis to a dashboard that supports faster, data-driven sales management decisions.